

Infrastructure audit

While you are there · Before departure, then first week on site

The status of every infrastructure requirement before Phase 1 goes live, with its evidence. Fill in before departure from what can be reported, then check each row on arrival.

Hospital	Unit / area	Completed by	Date
----------	-------------	--------------	------

1. The audit — status: met / partly / not met / unknown

Layer	Requirement	Minimum for Phase 1	How to verify	Status	Evidence
Power	HIS equipment never on unprotected supply	Servers and network core on UPS	Trace each server and core device to its supply		
Power	UPS energy from the critical load, with ageing factor	Time to shut down safely or bridge to the generator	Record the model and battery; measure runtime under load		
Power	Battery replacement interval for the room temperature	Interval known and budgeted	Log the room temperature; record installation dates		
Power	Supply voltage logged before choosing the UPS topology	One full day logged	Voltage logger at the supply point		
Power	Solar sized for the worst month and the autonomy required	Supply covers working hours	Survey panels, controller, batteries and loads; redo the battery sizing		
Location	Lockable cabinet, surge protection, room within the equipment's temperature range	Lockable cabinet; temperature within the servers' specification	Inspect the cabinet; log the room temperature		
Network	Every unit using the system reaches the server	Each Phase 1 unit connected	Test from each unit's workstation		
Network	Acceptable response time	Interactive response at every unit	Time typical actions in the system at the furthest units		
Network	Cabling within standard limits	Runs within limits	Survey the cabled sections		
Network	Segmentation with filtering	Documented, even if not yet implemented	Inspect router and switch configuration		
Server	Platform requirements	4 cores, 8 GB, 150 GB (Bahmni)	Inspect the hardware		
Server	A second machine able to take over	Backup server	Restore a virtual machine onto it		
Backup	3-2-1, one copy off site, encrypted	Daily backup; encrypted off-site copy	Check the schedule, a recent backup and who holds the key		
Recovery	Unattended	Shutdown on low	Cut the supply		

	recovery after a power cut	battery; power-on on return; start order set	under supervision and observe		
Recovery	Written downtime procedure; emergency paper forms; annual drill	Procedure written; forms for at least 8 h in each Phase 1 unit	Read the procedure; count the forms; run a drill		
Remote access	Outbound VPN through carrier NAT	Remote access to the server	Connect from outside; record who holds the keys		
Monitoring	Remote monitoring and alerts	An alert when the server stops	Stop a test service and confirm the alert		

Rows found unknown or not met are the first tasks of the trip. Requirements and sources: companion thesis §7.10.

2. Inputs — Electricity

Item	Finding	Source / photo
Grid hours/day (observed)		
Generator: fuel / capacity / manual or automatic start		
Solar: panels, controller, battery capacity, chemistry, installation date		
Voltage logged over a full day (min / max / typical)		
Outlet locations per building		

3. Inputs — Internet

Item	Finding	Source / photo
Provider and technology		
Contracted bandwidth		
Measured bandwidth (down / up, time of day)		
Carrier-grade NAT check: router WAN address vs public address		
Monthly cost		

4. Inputs — Space and environment

Item	Finding	Source / photo
Equipment room: ventilation / lock / keyholder / temperature		
Cable routes between buildings; anything crossing open ground or a road		
Distance between the farthest buildings; wall materials		
Humidity, dust, insects and rodents, flooding		
Lightning: frequency, existing protection		

5. Inputs — Existing IT

Item	Finding	Source / photo
Computers and tablets (count, OS, age)		
Printers		
Network equipment (models, where, working?)		

Existing cabling		
Existing software (finance, forms, earlier projects)		